

Workshop No. 2

Student: Laura Daniela Cubillos Escobar - 20211020045

① i) identifier = $r' \setminus b[a-z | A-Z | _][a-z | A-Z | _ | 0-9]^* \setminus b'$

21) integer = $r' \setminus b[0-9] + \setminus b'$

iii) $\text{float}(\text{literal}) = r' \setminus b[0-9] + \setminus . [0-9] + (?: [\text{Ee}] [+ -]? [0-9] +) ? \setminus b'$

iv) string literal = "1.2.3"

v) Singeline Comment = `/* % */`

```
vi) multilineComment = '^[\\s\\S]*\\*/'
```

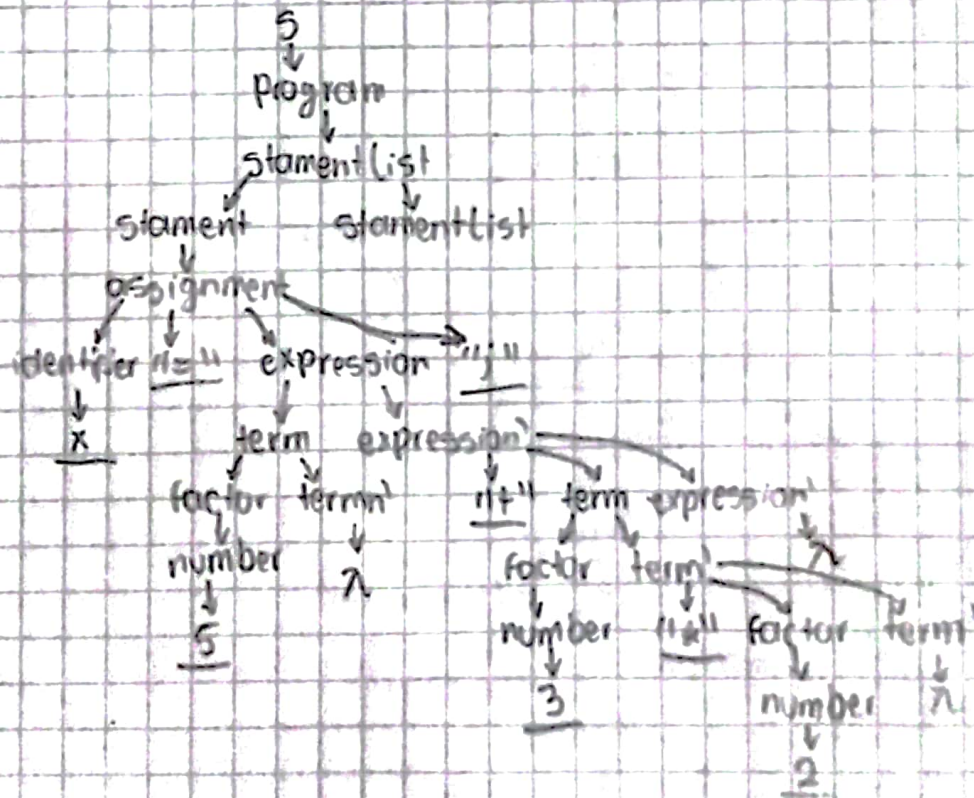
```
vi) whitespace = r'\s+'
```

vi) Operators = $r' \setminus b (= || = |A| - |A'| / |A|)$

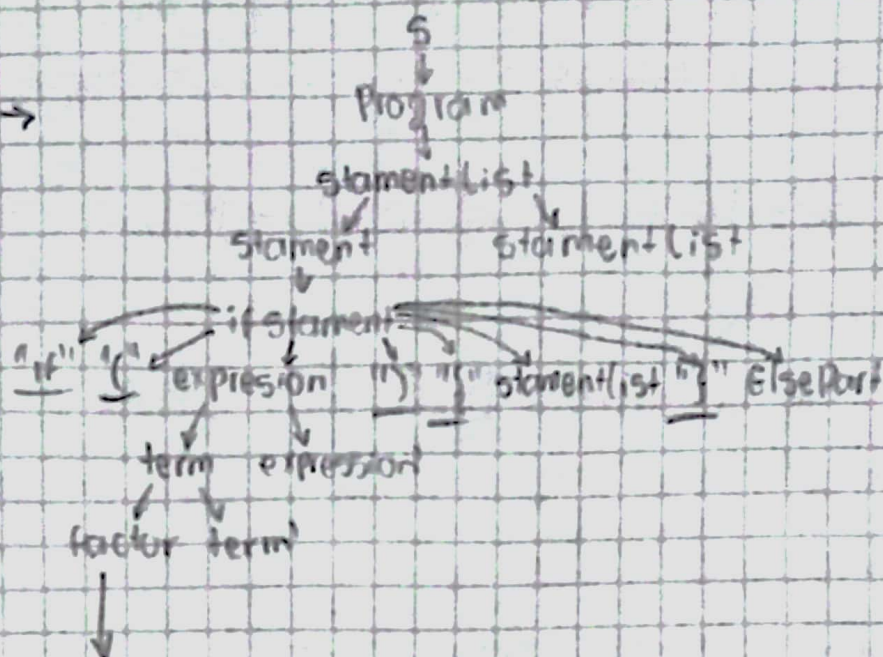
```
ix) keywords = r'\b(if|else|while|return)\b'
```

2) hexadecimal = r'\b0[xX][0-9a-fA-F]+\b'

② a) $x = 5 + 3 \cdot 2$



b) `if (x > 0) {
 y = x - 1;
} else {
 y = 0;
}`



In this point we can view that this statement can't be created by this grammar. There is not ">" expression.

c) `while (x < 10) {
 x = x + 1;
}`

→ As the previous point, this statement can't be created by this grammar. There is not "<" expression.

d) `return (a + b) * c;`

